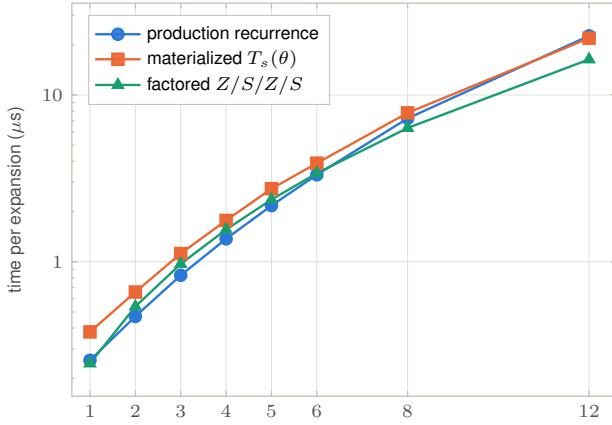
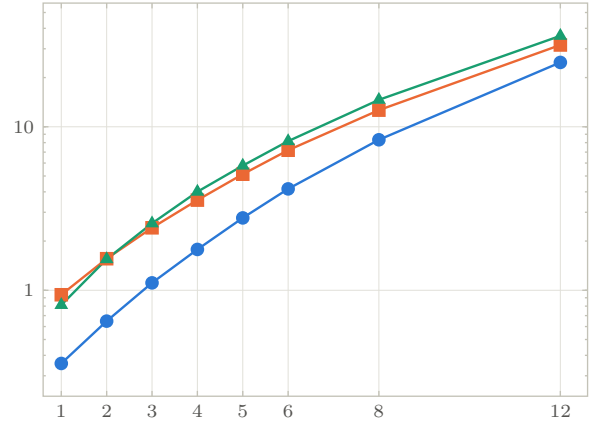


Fig. 1 — Crossover evidence: isolated per-column operators and the production whole-slab path

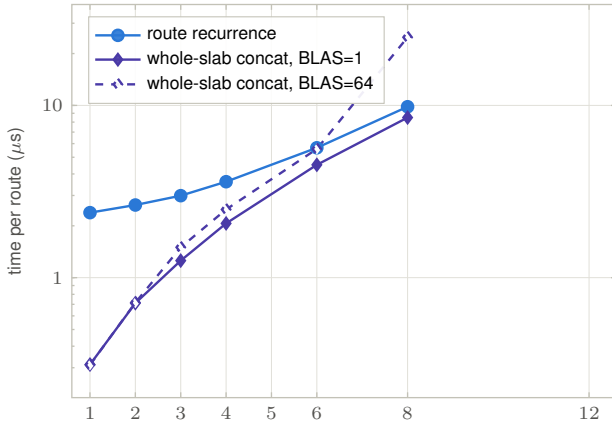
(a) φ only, BLAS=1



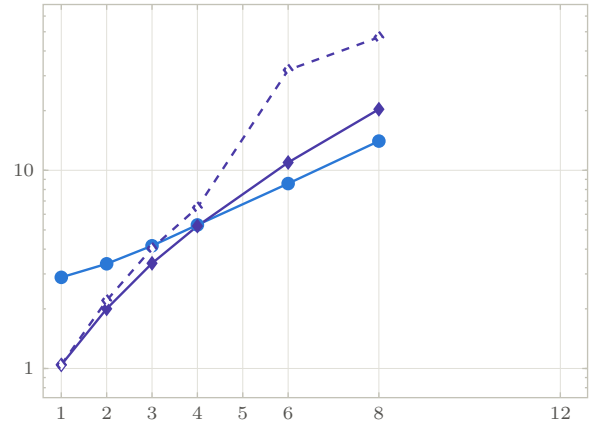
(b) Lamb–Helmholtz, BLAS=1



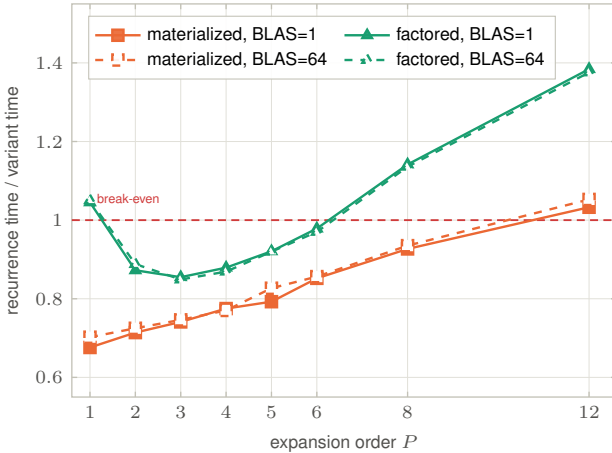
(c) whole slab, φ only



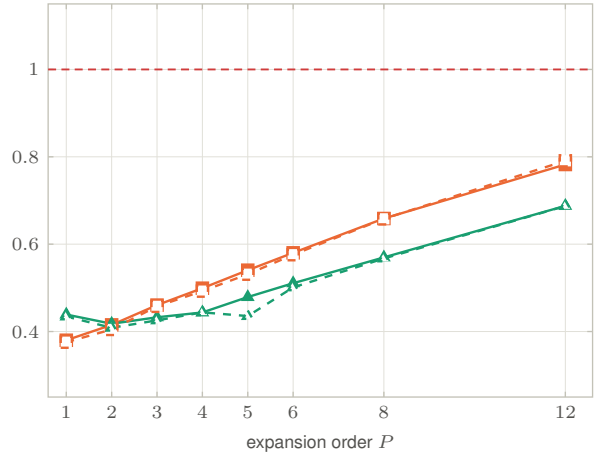
(d) whole slab, Lamb–Helmholtz



(e) per-column speedup, φ only



(f) per-column speedup, Lamb–Helmholtz



Machine / regime: m12-1-7, AMD EPYC 7763, Julia 1.11.7, OpenBLAS pinned at process start to 1 and 64 threads; Float64; batch = 64 expansions. **Data:** task 019b, data/smallp_fallback_layout/m12-1-7/crossover_{isolated,stage}_blas{1,64}.csv (tables fig01a_*, fig01b_*, fig01c_*). **Reads as:** the isolated per-column 014 operators are slower than the production recurrence at every $P \leq 6$ and in every BLAS regime; the materialized path only reaches parity near $P = 12$ and the factored path crosses break-even near $P = 8$ for φ -only, while for Lamb–Helmholtz neither variant ever wins in this range. In contrast, the whole-slab concat path already wins at small P , then approaches or crosses back through parity as order and the χ channel increase. **Caveats:** rows 1 and 3 are the isolated operator harness at batch 64; row 2 is the `tiny_parent` stage harness (3096 routes). They share a machine and per-expansion/per-route normalization but not harness scope, so they are deliberately adjacent rather than overlaid. The isolated kernels are scalar per-column, so per-expansion time is essentially batch-independent — the batch choice does not change the conclusion, and the 64-thread BLAS curves lie on top of the single-thread ones. Fig. 2 expands the whole-slab comparison across route counts and adds both the 019b corner and integrated 024 GPU-vs-host results.